

RAI Risk Taxonomy 2.0

Upgrade Technical Report

L4 cleaning, EM-based L3 mapping, and final release results

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English edition

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1 Executive summary

This project completed the operational pipeline for the RAI Risk Taxonomy 2.0 upgrade. The 892 source L4 records were frozen as an immutable baseline. Explicit instructions were then applied to deletion, consolidation, splitting, title and definition revision, and L1 routing. The cleaned total is 834 records, satisfying the reconciliation identity $892 - 39 - 20 + 1 = 834$.

Each L4 record was mapped only against the existing L3 categories within its confirmed L1 domain. The method was a seed-constrained spherical hard EM procedure. Korean and English embeddings were computed separately and combined with equal weight. Consensus across five initialisations produced 769 EM assignments. The remaining 65 boundary records were placed in Others and marked for human decision (HD), comprising 50 General, 2 Agentic, and 13 Physical records. No new formal L3 category was created, and all 46 source L3 rows remained unchanged at both file-hash and cell-content levels.

The five primary deliverables are `L1_Master.csv`, `L1_L2_L3_Master.csv`, `L4_General.csv`, `L4_Agentic.csv`, and `L4_Physical.csv`. External independent validation is a separate user-managed stage. This report records the completed internal processing, diagnostics, and validation.

2 Taxonomy and governing rules

The hierarchy comprises L0 Responsible AI, L1 General AI, Agentic AI, and Physical AI, L2 System, Interaction, and Societal Impact, L3 master categories, and L4 risk cards. The implemented sequence was:

1. freeze the sources and L3 master by hash and assign an immutable `source_row_id`;
2. archive explicit deletions, consolidate nominated duplicates, and split multi-meaning cards;
3. harmonise bilingual titles and definitions, giving priority to valid human-edited descriptions;
4. apply explicit L1 movement instructions;
5. run EM only over the existing L3 candidates within the confirmed L1; and
6. send low-margin, unstable, or bilingual boundary cases to Others and HD.

The L3 master took precedence whenever it conflicted with an Instruction Prompt. The single value in the second blank Physical header was ignored as instructed and remains only in the raw source CSV. `facet` and `act-type` were retained as L4 attributes but excluded from the semantic input.

3 Input data and integrity baseline

Table 1: Source and final L4 counts

Domain	Source L4	Final L4	Net change
General AI	591	618	+27
Agentic AI	83	85	+2
Physical AI	218	131	-87
Total	892	834	-58

The final domain counts reflect both structural cleaning and explicit L1 movement. Instruction Prompt parsing routed 103 active records, and the separately resolved Physical RAI4-0466 brought the instruction-based routing total to 104.

Table 2: Core input hashes

Input	SHA-256
L3 master CSV	b154be6ced881cc10d93779447afe3b636bca6094c399fc4d49076d3812c4

Input	SHA-256
General L4 CSV	97a57589bb218a89372bf703cb569c7e5344ff09155dfb74bc624b9a0bdeeb8
Agentic L4 CSV	d790d6feda006e0026d582aa0ad0b3d1e8442257eb563583e6077039ab4623cf
Physical L4 CSV	d578772484243a6888a2e41afbfb580385debbac2ca70c29c67a1fcebd349c
Harmonised input XLSX	39f36fa5ab31501a8738f59a3e71566d183d63aee4eca5eacaf188ab0931ec7a

4 L4 cleaning results

4.1 Deletion, consolidation, and splitting

All 39 explicit deletions came from the Physical source and were isolated from the active data. Fourteen consolidation groups removed 20 absorbed records. Their representatives received revised bilingual names and complete definitions. RAI4-0793 was separated into false information targeting individuals and the dissemination of security threats involving dangerous or sensitive information, creating one net additional record.

Table 3: L4 count reconciliation

Stage	Change	Remaining L4
Source	0	892
Explicit deletion	-39	853
Removal after consolidation	-20	833
Multi-meaning split	+1	834

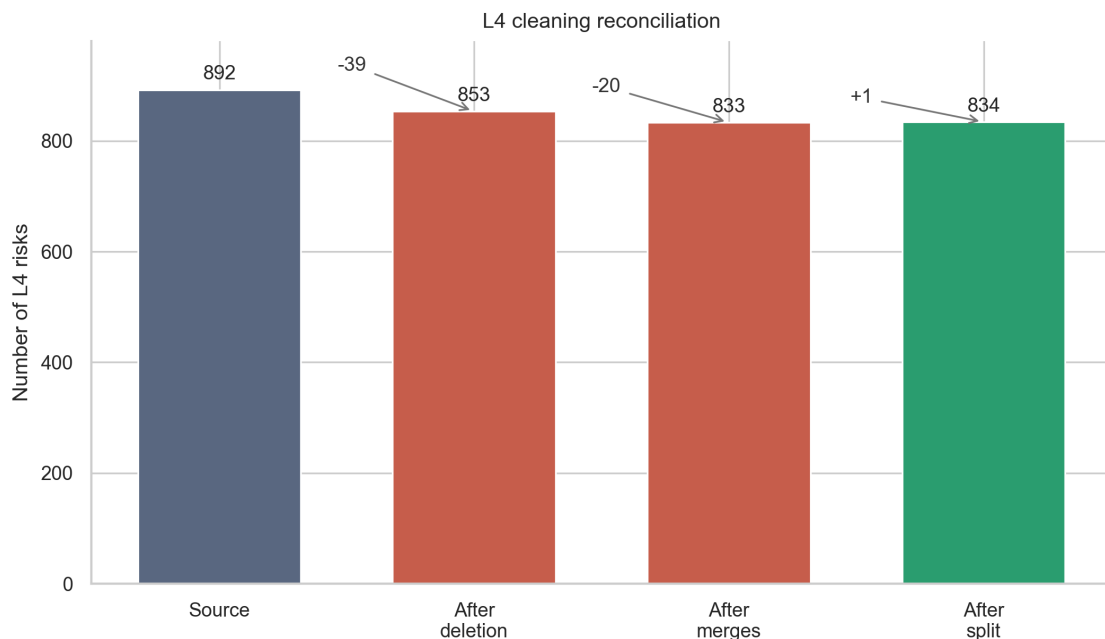


Figure 1: Count reconciliation across the L4 cleaning stages

Deleted source records are preserved in Deleted_Archive.csv, absorbed records in Merged_Archive.csv, and split ancestry in Split_Lineage.csv. ID_Crosswalk.csv links each original record to its final status and identifiers.

4.2 Bilingual content and attributes

All 834 final L4 records have non-empty Korean and English titles and definitions. Short extraction residue and standalone EAI tokens were removed. All 100 General source records carrying the old value new received a new L4 identifier. Agentic and Physical facet and act-type remain blank because their sources supplied no values.

5 L1 routing and identifier design

After explicit movement instructions, General contains 618 records, Agentic 85, and Physical 131. The principal reason for the fall in Physical was the transfer of general-model and software-agent risks into General and Agentic.

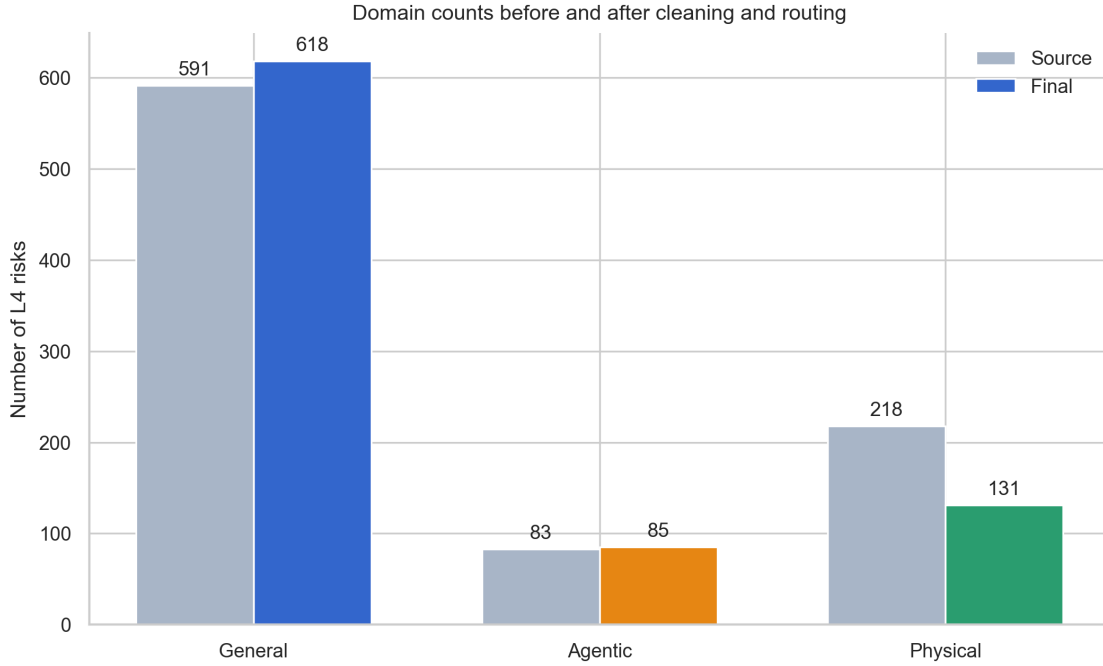


Figure 2: Domain counts before and after cleaning and L1 routing

L1 identifiers are L1_G, L1_A, and L1_P. L2 and L3 identifiers encode their parent hierarchy and distinctive meaning. Every L4 identifier was regenerated as L3_ID + sequence; for example, P_SYS_CONTROL_001 is the first L4 under Physical, System, and Unsafe Physical Control and Actuation.

Others is a derived HD queue, not a modification of the formal L3 master. Its identifiers are G_Others, A_Others, and P_Others. L2 fields are blank for these rows so that no new L2 structure is implied.

6 EM mapping method

6.1 Representation and E-step

The pipeline loaded the configuration, tokenizer, and `pytorch_model.bin` from one coherent local BAAI/BGE-M3 snapshot. It used the CLS vector followed by L2 normalisation. Files from different snapshots were not mixed. Only the final title and final description entered the embedding. Instruction Prompts, prior human L3 assignments, facet, and act-type were excluded.

For record i and candidate L3 k , the joint score is

$$s_{ik} = \frac{1}{2} \cos(\mathbf{x}_i^{ko}, \boldsymbol{\mu}_k^{ko}) + \frac{1}{2} \cos(\mathbf{x}_i^{en}, \boldsymbol{\mu}_k^{en}). \quad (1)$$

The E-step computes $z_i = \arg \max_k s_{ik}$ only within the confirmed L1 domain.

6.2 M-step and repeated initialisation

The M-step retains each L3 master embedding as an anchor and combines it with the mean of the assigned L4 vectors:

$$\mu_k^{(t+1)} = \text{norm} \left(2\mathbf{a}_k + \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i \right). \quad (2)$$

An empty cluster returns to its original L3 anchor. Five initialisations used seeds 20260826 to 20260830. The final result is the consensus assignment with a per-record stability score. The implementation is therefore a seed-constrained spherical hard EM procedure rather than a full probabilistic mixture-model EM.

6.3 Others and HD rejection rule

The operational boundary signals were a top joint score below 0.30, a top-two margin below 0.005, or stability below 0.60. A Korean-English Top-1 disagreement triggered HD only when the joint margin was below 0.008. This avoided treating ordinary cross-language phrasing differences as automatic rejection. One explicit request for an L3 absent from the Physical master was also routed to HD.

These are internal operational reject rules. Independent calibration against a stratified expert-labelled sample was outside the scope of this run.

7 Mapping results

Table 4: EM and HD results by domain

Domain	Final L4	EM	HD/Others	HD share
General AI	618	568	50	8.1%
Agentic AI	85	83	2	2.4%
Physical AI	131	118	13	9.9%
Total	834	769	65	7.8%

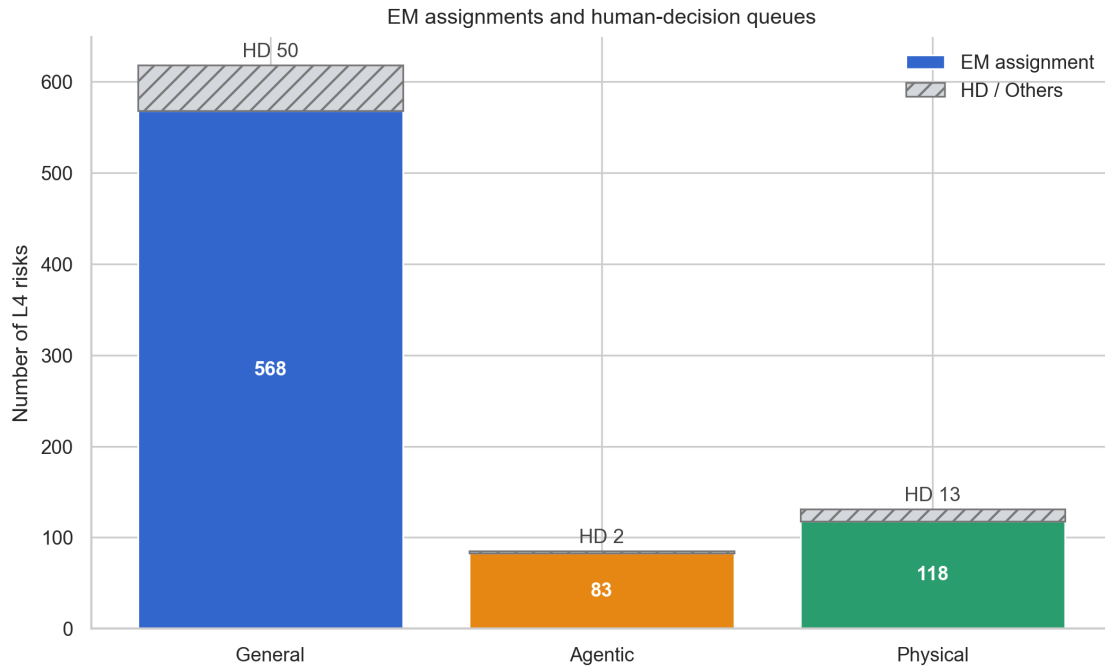


Figure 3: EM assignments and HD/Others queues by domain

Across the five runs, convergence required 5 to 7 iterations for General, 2 to 3 for Agentic, and 4 to 5 for Physical. The final objective ranges were 0.739425 to 0.739750, 0.754682 to 0.754912, and 0.725065 to 0.725349 respectively.

Table 5: EM quality diagnostics

Domain	Mean top score	Median margin	Mean stability	Bilingual Top-1 agreement
General AI	0.7395	0.0289	0.9686	70.7%
Agentic AI	0.7547	0.0357	0.9741	83.5%
Physical AI	0.7252	0.0247	0.9786	75.6%

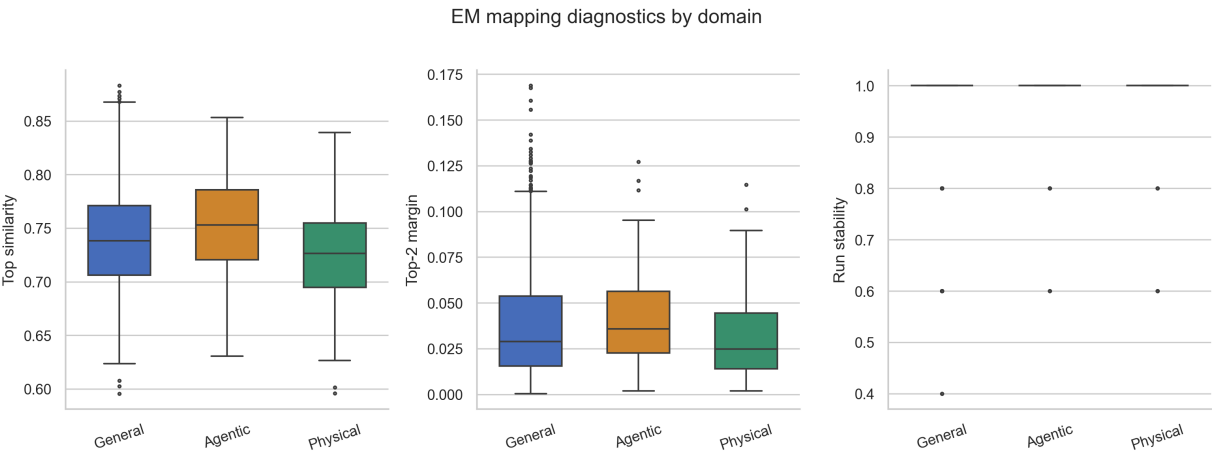


Figure 4: Distributions of top similarity, top-two margin, and run stability

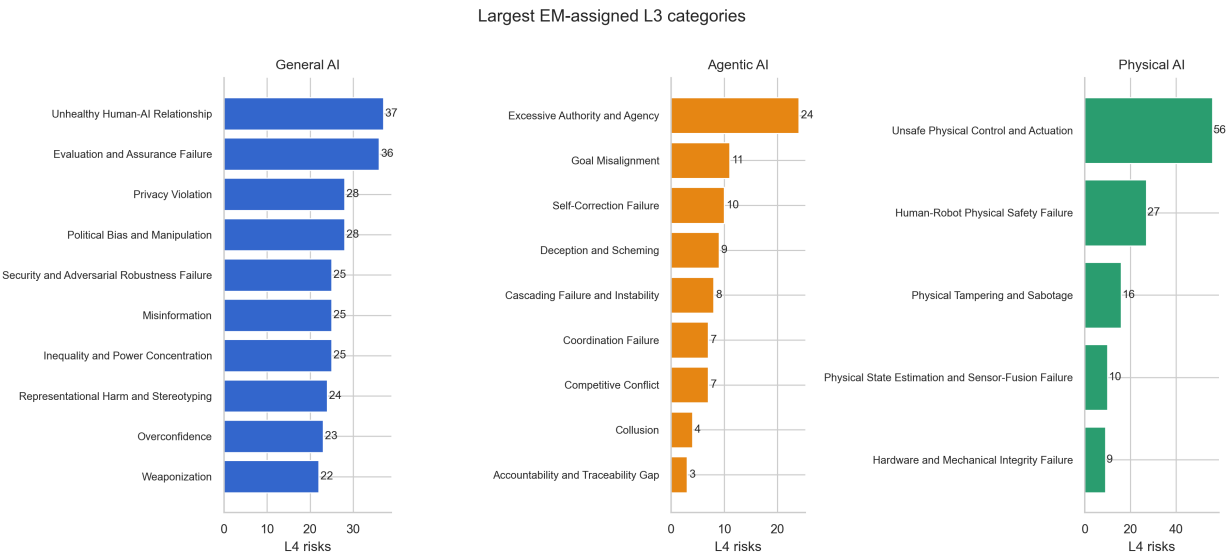


Figure 5: Largest EM-assigned L3 categories within each domain

8 Three internal validation methods

8.1 1. Source reconciliation and conservation

SHA-256 hashes were compared before and after processing, and the 46-row L3 master was re-read for equality. The cleaning conservation identity was checked against the final 834 records. Every source row was required to appear in `ID_Crosswalk.csv` as deleted, absorbed, active, or split. This method detects data loss, residual duplicates, and unintended L3 modification.

8.2 2. Convergence and initialisation stability

Five initialisations were run for each domain. The pipeline retained objective values, iteration counts, and per-card assignment agreement. Mean stability exceeded 0.96 in every domain and the final objective ranges were narrow. This method detects assignments that depend strongly on a particular initialisation.

8.3 3. Bilingual cross-checking and boundary isolation

Korean and English were embedded separately. The pipeline stored language-specific Top-1 L3 categories, joint similarity, and the top-two margin. Records with combined bilingual ambiguity or other boundary signals were placed in Others with an explicit `HD_Reason`. This reduces over-confident assignment caused by translation differences, multiple meanings, or inadequate L3 coverage.

All ten automated QA checks passed. They covered source-hash preservation, the 46 immutable L3 rows, L4 reconciliation, unique final IDs, complete bilingual fields, valid EM/HD labels, HD-only use of Others, valid L3 references, and the presence of exactly five primary CSV files.

9 Final release structure

Table 6: Primary release CSV files

File	Rows	Role	Main content
<code>L1_Master.csv</code>	3	L1 master	L0 and L1 IDs, bilingual names and definitions
<code>L1_L2_L3_Master.csv</code>	49	hierarchy	46 immutable L3 rows plus 3 derived Others rows
<code>L4_General.csv</code>	618	General	full L0 to L4 hierarchy, attributes, EM/HD, lineage
<code>L4_Agentic.csv</code>	85	Agentic	full L0 to L4 hierarchy, attributes, EM/HD, lineage
<code>L4_Physical.csv</code>	131	Physical	full L0 to L4 hierarchy, attributes, EM/HD, lineage

Each L4 file contains the IDs, Korean and English risk names, and Korean and English definitions for L0, L1, L2, L3, and L4. It also contains Others information, `facet`, `act-type`, EM score, margin, stability, language-specific Top-1 results, the HD reason, source identifiers and instructions, and transformation lineage.

10 Limitations and conclusion

This is an operational semantic mapping to the supplied L3 master. Its theoretical completeness, final decisions for the 65 HD records, and expert calibration of the rejection thresholds remain matters for external independent validation. Physical has only five source L3 categories, so some societal-impact or fairness risks can legitimately remain in Others. The pipeline completed cleaning, routing, EM mapping, identifier issuance, five-file release, audit logging, and visualisation without changing any source or L3 master file. The final reproducible result is 834 cards, comprising 769 EM assignments and 65 HD records.